Week40- Assignment-Solution

Problem 1 - Prototyping the physical design

a) Explain how CNC milling works and state the benefits of CNC milling for designing prototypes on the physical design.

CNC milling is reminiscent of 3D printing, but is a subtractive production - that is, starting with a subject and removing unwanted parts. A bit of everyone like a sculptor works.

Since it is much easier to cut than assemble, CNC cutters can work with far more materials than 3D printers - steel (for industrial machines), wax, plastic, wood and aluminium are among the most common materials.

b) Explain how laser cutting works and state the benefits of laser cutting for designing prototypes on the physical design.

The laser cutter is not as well-known as the 3D printer, but can often be a useful tool: It is easier to come up with good designs for the laser cutter.

The laser cutter takes a wide range of materials, among other things: Felt, leather, cloth, cardboard, MDF and wood.

Laser cutters are available both as industrial models and in less expensive models for home use.

Laser cutting works by controlling the x and y directions of the laser beam, as well as the power of the laser beam - In this way you can both etch and cut the materials.

By cutting thin strips through relatively rigid materials, they can be made flexible.

c) Explain how 3D printing works, name three different ways to do 3D printing, and state the advantages and disadvantages of 3D printing for designing prototypes on the physical design.

3D printing is a form for additive production: you start with nothing and build something together; and unlike laser cutting, which is subtractive production: You start with a subject and remove parts of it.

One of the great advantages of 3D printing is that you can make parts that are not possible with the other methods - For example, you can build continuous rings or joints without joints.

Task 2: Pathfinding

Use the online solution found here: https://clementmihailescu.github.io/Pathfinding-Visualizer/# and show the search algorithms we have been through- Depth First Search, Breadth first, Dijkstra, Best first, and A * by taking screenshots. Try to set up the path so that the unique with each algorithm is displayed. Give a brief description of what we see and how this differentiates the algorithm from the others.

Task 3: Pathfinding – manual tracing

Given the weighted graph below, what is the optimal path from a start node A to a target node J? Hand-trace and show your work for Depth First, Breadth First, Dijkstra, Best First, and A * search algorithms.

